

- 6 Fig. 6.1 is a diagram showing the passage of water through the tissues of a flowering plant from the soil to the atmosphere. The arrows show the direction of water movement.

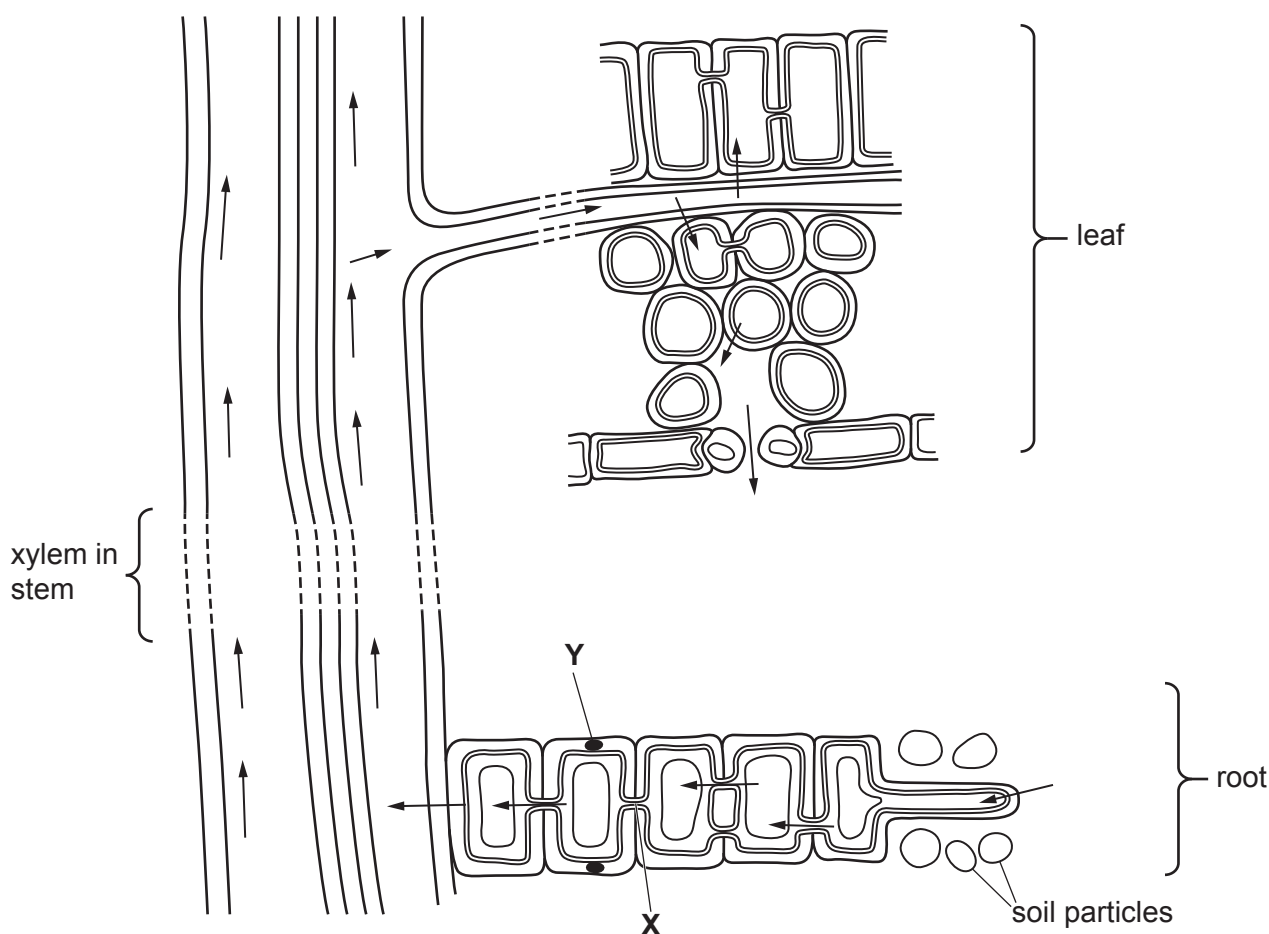


Fig. 6.1

- (a) The structure labelled **X** is part of the symplast pathway. State the name of structure **X**.
 [1]
- (b) The structure labelled **Y** in the cell wall is a barrier to the apoplast pathway. State the name of structure **Y**.
 [1]
- (c) With reference to Fig. 6.1, complete the statements about the movement of water in the flowering plant.
- Water moves from the soil solution to the cytoplasm of root hair cells by
- Water moves from the xylem in the root to the leaf by
- Water moves from mesophyll cell walls to intercellular air spaces by
- Water vapour moves from intercellular air spaces to the atmosphere outside the leaf by

[4]

[Total: 6]